

Quantum Leaps: A Computational Tour of Ten BPhO Tasks

Jermaine Antwi

Kingsford Community School, London, United Kingdom

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Acknowledgements

My thanks go to the British Physics Olympiad for setting a challenge that rewards building something rather than just answering something, and to G-Research for sponsoring it. Nearly every formula here comes from the Challenge Presentation pack or the Quantum Mechanics problem sheets that came with it. The sheets did more work than I expected: two of their questions corrected how I had read the main task briefs, and rather than quietly fixing those I have written up what I had misunderstood in Section 14, because those were the two places I learnt the most.

Thanks also to my science teacher, Mr Galagedera, for always encouraging me to challenge myself and to go looking for harder problems outside the curriculum rather than stopping at what was set. This project, and the extension work in particular, is a direct result of that.

This project leans on a lot of open-source software. The numbers and figures come from NumPy, SciPy and Matplotlib; the app is built with React, TypeScript, Vite, Tailwind CSS, Recharts and three.js. This document is typeset in $\text{\LaTeX}$ and compiled with Tectonic. I am grateful to everyone who maintains them.

Abstract

This project presents ten computational models developed for the BPhO Computational Physics Challenge 2026, focusing on quantum mechanics. The tasks cover a range of phenomena from introductory topics like random walks and Brownian motion to more advanced concepts including electron diffraction, entangled photons, and hydrogenic orbitals.

The core of the submission is an interactive web application that visualises these physical systems in real time. To ensure the calculations in the app were reliable, I also implemented parallel Python scripts to generate the static plots for this write-up. Comparing the outputs from both languages served as the primary debugging method throughout the project.

In addition to the core tasks, this write-up covers four extensions. These include an attempt to derive the uncertainty product for a particle in a box, alongside an evaluation of the CHSH inequality. It also contains a look into superpositions of hydrogenic orbitals, and full solutions to problem sheets QM1 to QM3. The results throughout are validated against standard formulas and expected theoretical limits.

Contents

Acknowledgements	2
1 Introduction	6
1.1 How I built it	6
1.2 On checking things	6
1.3 Software	6
1.4 Structure of the report	7
2 The Random Walk	7
3 Brownian Motion	7
3.1 Adding the walls and fixing speeds	9
4 Blackbody Radiation and the Einstein Solid	10
4.1 The catastrophe	10
4.2 The Einstein solid	10
4.3 Checking against Stefan-Boltzmann	10
5 Task 4: Shedding Light on the Matter — the Photoelectric Effect	11
5.1 Getting Planck’s constant back out	11
6 The Hydrogen Spectrum and the Bohr Model	11
6.1 The reduced mass correction	12
7 Electron Diffraction	13
7.1 Avoiding a circular check	14
8 The Particle in a Box	14
8.1 Extension: the uncertainty principle	15
9 Quantum Cryptography and the Bell Test	15
9.1 Extension: CHSH	16
9.2 Simulation	16
10 Compton Scattering	16
10.1 Visualizing the recoil	17
11 Hydrogenic Orbitals	18
11.1 Rendering challenges	18
12 The Uncertainty Principle for a Particle in a Box	18
12.1 Position uncertainty	18
12.2 Momentum uncertainty	20

12.3 The product	20
13 Hybrid and Molecular Orbitals	20
13.1 Hybrid orbitals, and where bond angles come from	21
13.2 Molecular orbitals by LCAO	21
13.3 Morphing between m states	22
14 Problem Sheets QM1–QM3	23
14.1 QM1: scales, radiation and the photoelectric effect	24
14.2 QM2: photons, relativity and stars	25
14.3 QM3: uncertainty, confinement and entanglement	26
15 Validation	27
15.1 Deterministic checks	28
15.2 Stochastic checks	28
15.3 Cross-implementation agreement	29
16 The Interactive Application	31
16.1 Introduction	31
16.2 Libraries and technologies	31
16.3 Architecture	31
16.4 Functionality	31
16.5 Implementation notes	32
16.6 Deployment	32
16.7 Summary and possible improvements	32
16.8 Links	33
17 Conclusion	33
17.1 A few surprises	33
17.2 Takeaways from the project	33
17.3 Future work	33

1 Introduction

Quantum mechanics is what you get when you follow a handful of experiments that classical physics simply cannot explain. The spectrum of a hot object, the photoelectric effect, hydrogen's discrete emission lines, electrons producing diffraction rings. Every one of those is a task in this challenge, and taken in order they walk through roughly the route the theory was actually assembled along, from Planck quantising radiation in 1901 [20] to Bell's theorem [2] and the experiments that settled it [1].

This paper covers all ten tasks and four extensions, and I've tried to show my working wherever a derivation is manageable. For the advanced math (like the spherical harmonics), I just used the formulas given in the Challenge Presentation pack [5] or looked them up in textbooks, as the brief permits.

1.1 How I built it

I made two main choices at the start of the project.

The first is that each task is a slide in one interactive app rather than a separate script. Every number on screen is recomputed from the controls the moment you move them. Building it this way was harder than just writing separate scripts, but it helped me understand the physics a lot better when I could actually see things change in real time.

The second is that I built all of it twice, once in TypeScript for the app and once in Python for the figures in this document. That started out as a way of avoiding a write-up made of screenshots. It turned into a helpful debugging tool. The two implementations are completely separate: for example, I used `scipy.special` for the hydrogen orbitals in Python, but had to figure out how to evaluate the polynomials manually in TypeScript using a recurrence relation from a textbook. Because they share no code, getting them to agree meant they were probably both right. Several bugs described in this paper only got caught because the Python and TypeScript outputs disagreed slightly.

1.2 On checking things

The tricky bugs were rarely the ones that crashed the app. Instead, a figure would come out smooth and plausible but the physics underneath it was wrong, and there is no way to catch that by just looking harder at the picture.

Most task sections here try to outline what the model is, the issues I ran into, and the checks I used to verify the output. Section 15 pulls all of those checks together, and separates the deterministic ones from the Monte Carlo ones, because those two need completely different standards of evidence.

1.3 Software

The numerical work uses NumPy [14], SciPy [25] for special functions, and Matplotlib [16]. The app is React [17] and TypeScript [18], bundled by Vite [26], styled with Tailwind CSS [23], with Recharts [21] for the 2D charts and three.js [6] through react-three-fiber for the orbital rendering. Constants are CODATA 2018 [24], except where a problem sheet prints its own, in which case I use the sheet's numbers so its answers reproduce.

1.4 Structure of the report

Sections 2 to 11 are the ten tasks in order. Sections 12 to 14 are the extension work: the uncertainty principle, superpositions of orbitals, and the problem sheets. Section 15 is the validation. Section 16 documents how the app is built. Section 17 concludes.

2 The Random Walk

Model a random walk of N steps of fixed size s , each step in a direction drawn uniformly from $[0, 2\pi)$. First task, and it sounds easy enough.

My first version actually picked a random x offset and a random y offset independently. This turned out to be wrong, but it was hard to spot at first. Picking the two components independently makes every square patch of the plane equally likely, rather than every direction equally likely. The diagonals end up being longer than the axes by a factor of $\sqrt{2}$. As a result, the walk fills a square-cornered blob instead of a circular one. Since a single trajectory looks mostly fine, I only noticed the issue when I plotted fifty of them on top of each other.

The fix was just to sample the angle instead of the components:

$$x_{i+1} = x_i + s \cos \theta, \quad y_{i+1} = y_i + s \sin \theta. \quad (1)$$

Now the symmetry is built in rather than hoped for.

The brief asks you to check that the mean square displacement scales as Ns^2 , which falls out of the steps being independent: write $\mathbf{r}_N = \sum_i \mathbf{s}_i$, and every cross term $\langle \mathbf{s}_i \cdot \mathbf{s}_j \rangle$ with $i \neq j$ averages to zero because the directions are uncorrelated, leaving N diagonal terms of s^2 each. Averaging r^2 over 120 walks at $N = 600$, $s = 1$ gave me 355.1 against a predicted 399.0.

I spent a while trying to find the bug that caused that 11% gap, but there wasn't one. I was just averaging a random variable over a finite number of samples and treating the answer as an exact measurement. The more relevant question is whether the error shrinks like $M^{-1/2}$ as you add samples (Section 15 shows it does across three decades). Working out which of my numbers were exact and which were noisy was a bit of a learning curve, and it mattered for Tasks 2 and 8 as well.

3 Brownian Motion

N small particles of mass m and radius r moving randomly, some of them colliding with a large particle of mass M and radius R , the small ones modelled as random walks and the collisions handled by conservation of momentum.

The Brownian motion model treats the jitters of a large particle as the macroscopic result of discrete collisions with smaller background particles. Einstein's 1905 treatment [13] formalized this mechanism.

The simulation of the small background particles simply reuses the random-walk logic from Task 1. For the collisions, the pack says to reflect both velocities about the centre-of-mass velocity, with a restitution coefficient C setting how much energy is lost. The reason this conserves momentum is worth spelling out, because if you move into the frame travelling with the pair's centre of mass, the two momenta are equal and opposite by definition, so the total is centre of mass, the two momenta are equal and opposite by definition, meaning their vector sum is zero. Reflecting both velocities preserves this zero total, guaranteeing momentum conservation when transformed back to the laboratory frame. Setting the restitution coefficient $C < 1$ scales down the relative speed after collision to simulate inelastic energy loss.

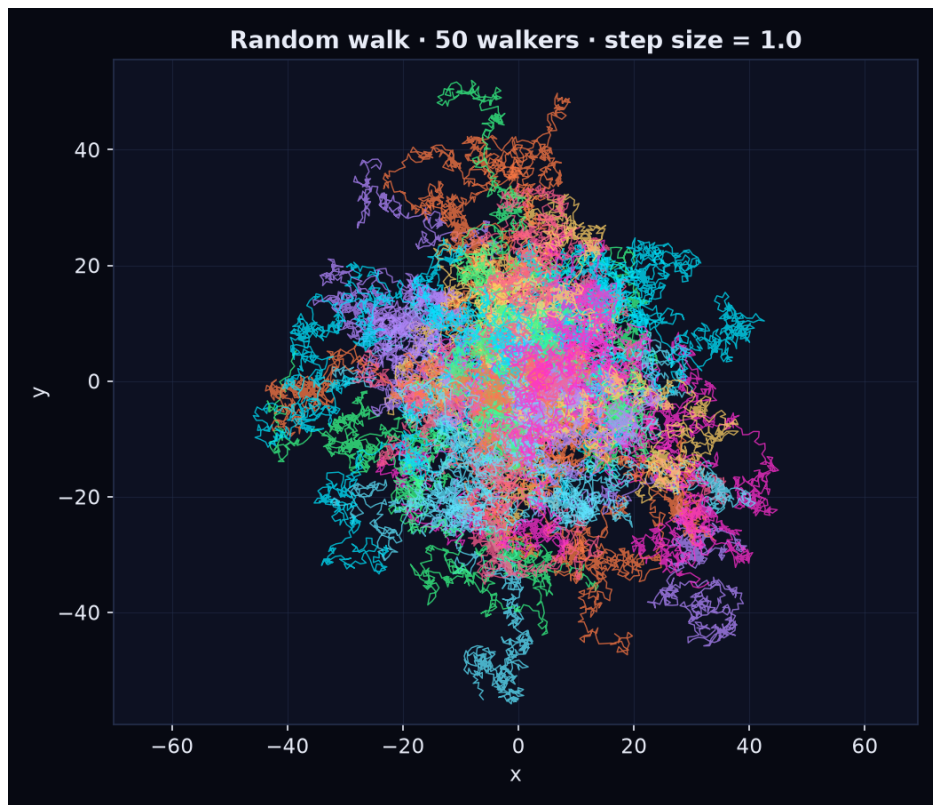


Figure 1: Fifty walks overlaid. The distribution is circular, which is exactly what the earlier component-sampling version was not, and overlaying them is the only way I could see the difference.

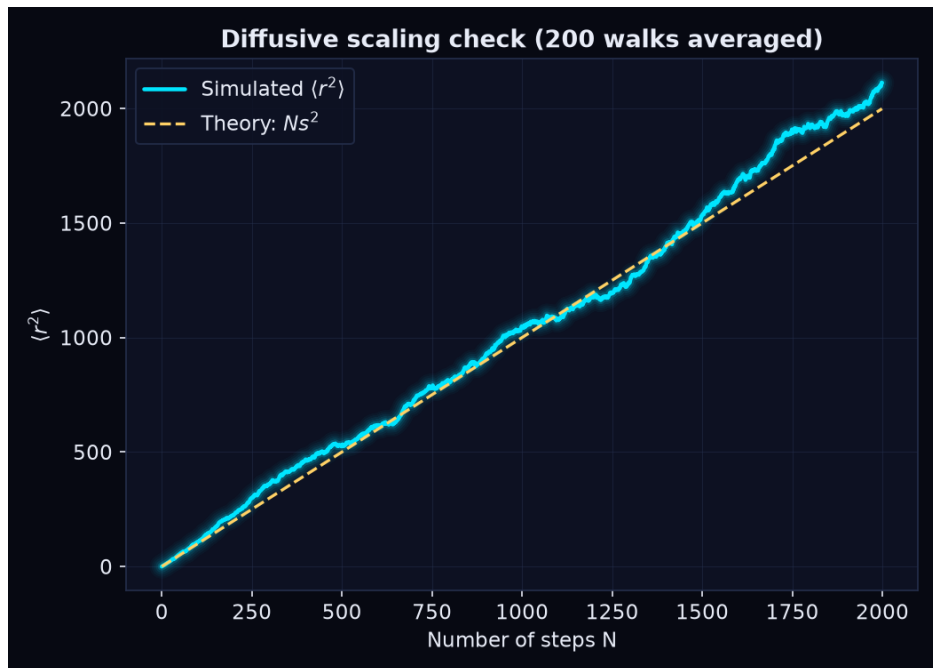


Figure 2: $\langle r^2 \rangle$ averaged over many walks against the Ns^2 prediction. It bounces around the line and tightens onto it the more walks you throw at it.

3.1 Adding the walls and fixing speeds

Particles drifting out of the simulation bounds requires implementing a rigid-wall boundary condition. A simple position-reversal algorithm reflects any particle (background or tracked) that attempts to step beyond the assigned simulation dimensions, preserving its speed while reversing the relevant velocity component.

Another important detail to get right in the simulation is the initial speed, since the standard formula $v = \sqrt{3k_B T/m}$ is for 3D space, where equipartition gives each independent direction $\frac{1}{2}k_B T$ on average. In a 2D simulation, there are only two directions to carry energy, so:

$$\frac{1}{2}mv^2 = k_B T \implies v = \sqrt{\frac{2k_B T}{m}}. \quad (2)$$

Using the 3D equipartition formula inadvertently scales up all initial speeds by a factor of $\sqrt{3/2} \approx 1.22$.

The extension provides a slider for the restitution coefficient. Dragging C below 1 visibly damps the large particle's motion as kinetic energy drains from the system during each inelastic collision.

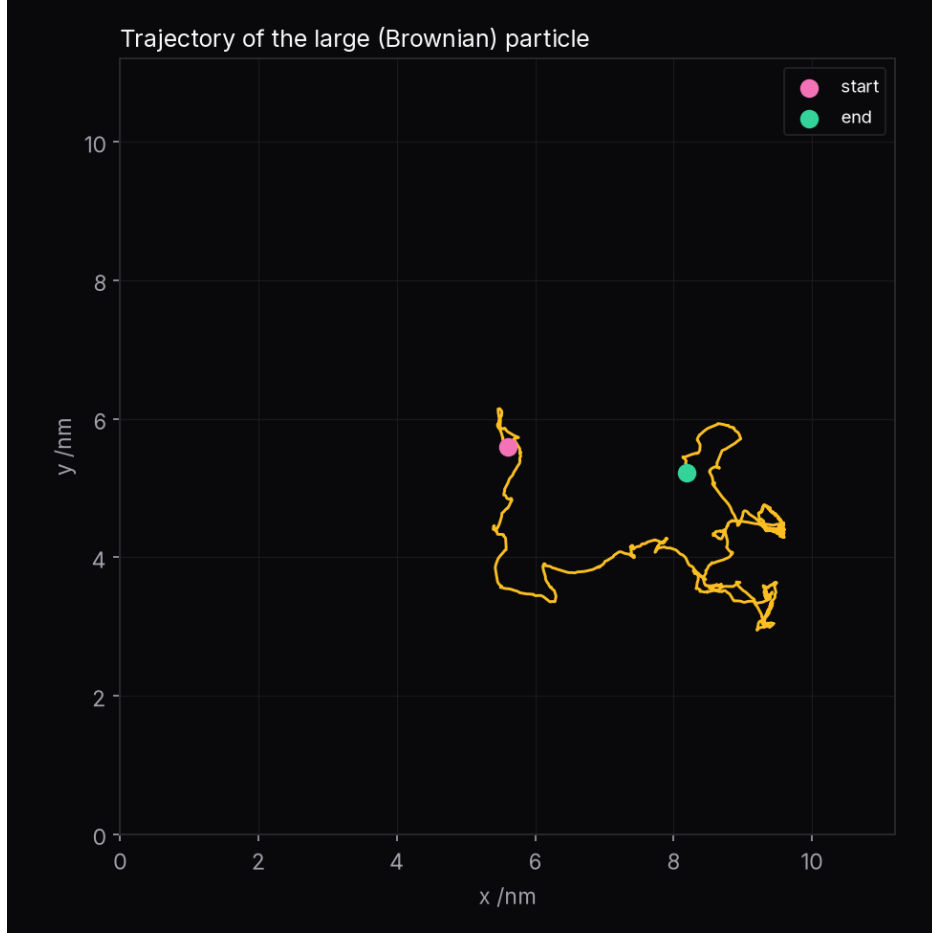


Figure 3: The tracked particle's path bounded by specular wall reflections.

4 Blackbody Radiation and the Einstein Solid

Plot the Planck spectrum $B(\lambda, T)$ at several temperatures, and Einstein's model of molar heat capacity against temperature for atomic crystals.

4.1 The catastrophe

The interesting bit is what happens at short wavelength. Treat light as a classical continuous wave and you get the Rayleigh–Jeans law, $B \propto \lambda^{-4}$, which runs away to infinity as $\lambda \rightarrow 0$. The divergence of the classical Rayleigh–Jeans law alongside Planck's distribution illustrates the ultraviolet catastrophe. Planck's fix [20] was to quantise the field in lumps of hf , which cuts the short-wavelength modes off:

$$B(\lambda, T) = \frac{2hc^2}{\lambda^5} \cdot \frac{1}{\exp\left(\frac{hc}{\lambda k_B T}\right) - 1}. \quad (3)$$

I plot both, clipping the classical curve because otherwise it flattens everything else into the axis. The curves converge at long wavelengths and diverge at short wavelengths, reflecting the impact of the quantisation correction as hf approaches $k_B T$.

4.2 The Einstein solid

Same idea, applied to atoms vibrating in a crystal:

$$C_v = 3R \frac{x^2 e^x}{(e^x - 1)^2}, \quad x = \frac{hf_E}{k_B T}. \quad (4)$$

The limit at high temperatures is $3R$, known as the Dulong–Petit law, while the capacity collapses to zero at low temperatures.

Diamond needs a much higher temperature than gold to reach the flat part because its atoms vibrate at a far higher frequency, meaning more thermal energy is needed to excite even the first vibrational mode.

4.3 Checking against Stefan-Boltzmann

To verify the implementation, Equation 3 is integrated numerically over all wavelengths and compared against the Stefan–Boltzmann law. Because σ does not appear in Planck's formula, it must emerge from the integral.

B is a spectral radiance (power per area per solid angle), whereas σT^4 is the total emitted radiant emittance. Integrating over the hemisphere and weighting by $\cos \theta$ accounts for the projected surface area:

$$\int_{\text{hemisphere}} \cos \theta \, d\Omega = \int_0^{2\pi} \int_0^{\pi/2} \cos \theta \sin \theta \, d\theta \, d\phi = \pi, \quad (5)$$

not the 4π of a whole sphere. So $\sigma T^4 = \pi \int_0^\infty B \, d\lambda$.

Between 3000 and 9940 K the two agree to better than 0.0003% (Table 1).

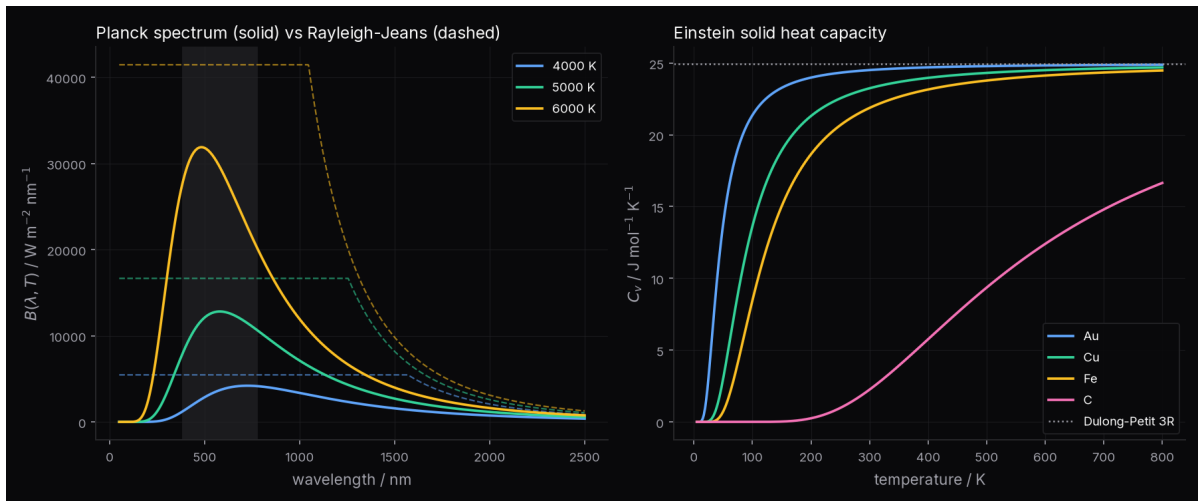


Figure 4: Left: Planck at several temperatures against the classical prediction, which diverges. Right: Einstein heat capacity for several elements, heading for the Dulong–Petit limit of $3R$.

5 Task 4: Shedding Light on the Matter — the Photoelectric Effect

If light is a wave, then brighter light should knock electrons out harder. That was my assumption and it is wrong. Turning the intensity up gives you more electrons, not faster ones, and below a threshold frequency you get nothing at all no matter how bright the source is.

The only picture that survives that, and the one Einstein got the Nobel Prize for [12], is that light arrives in packets of energy hf . One packet either has enough to free an electron or it does not, and how many packets per second are turning up is irrelevant to that question. The pack writes it as

$$eV_{\text{stop}} = hf - W, \quad (6)$$

where V_{stop} is the voltage that just stops the fastest electrons and W is the work function, the energy an electron loses simply getting out of the metal before it is free to go anywhere.

5.1 Getting Planck’s constant back out

Equation 6 is a straight line in f with gradient h/e (how h was apparently first measured directly). That is too good to leave alone, so rather than just plotting the line I treated it as a measurement: twelve simulated stopping voltages for a metal whose work function I chose, 20 mV of Gaussian noise thrown on top to stand in for instrument error, then a straight-line fit, then read h off the gradient and W off the intercept.

It gives $h = 6.640 \times 10^{-34} \text{ Js}$ against the real $6.626 \times 10^{-34} \text{ Js}$, out by 0.20%, and $W = 2.312 \text{ eV}$ against the 2.300 eV I put in, out by 0.51%. Both are about what the injected noise should give you. For twelve noisy points I was fairly pleased with that, and more to the point it tests the gradient rather than just the shape.

6 The Hydrogen Spectrum and the Bohr Model

Graph photon energy against wavelength for emissions from hydrogen due to transitions between levels, with Z variable.

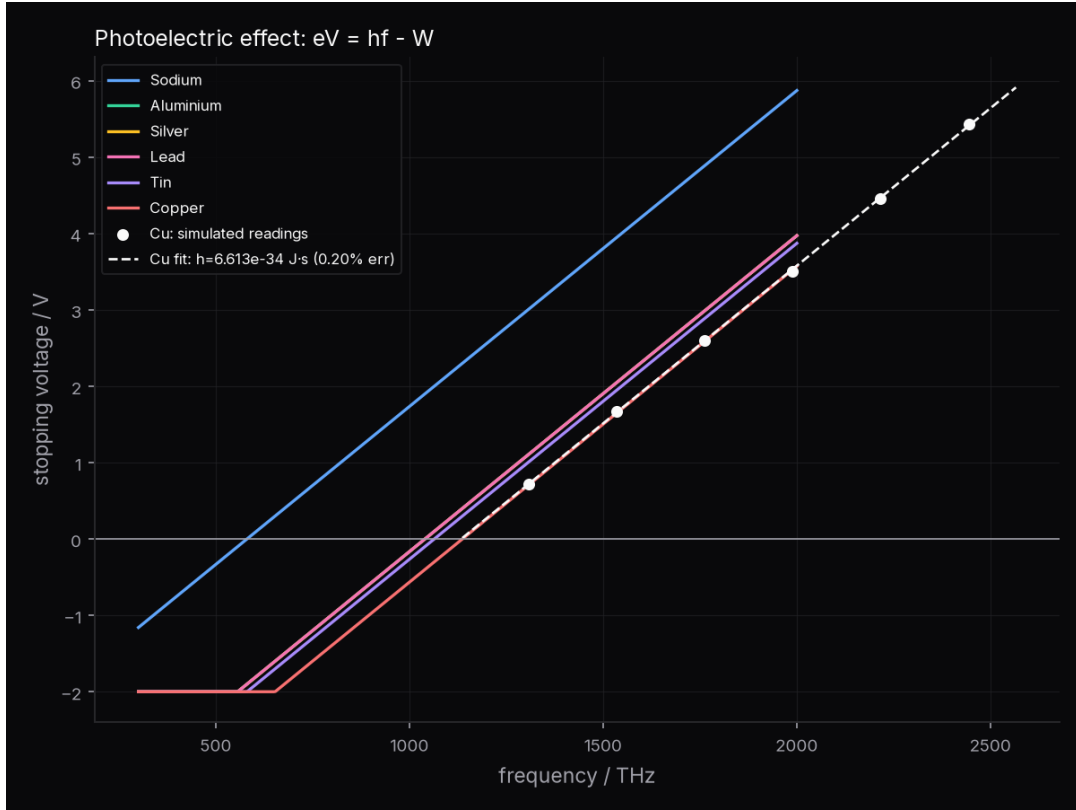


Figure 5: Stopping voltage against frequency for six metals. The lines are parallel because they all share the gradient h/e and differ only in intercept, the work function.

The planetary model of the atom is classically unstable because an accelerating electron radiates energy and spirals into the nucleus. Bohr [4] bypassed this by asserting that only specific orbits are permitted, where angular momentum is an integer multiple of $\hbar$, and that electrons in these orbits do not radiate. The physical justification for this quantization condition was later provided by de Broglie [11]. If the electron is a wave of wavelength $\lambda = h/p$ wrapped around a circular orbit, it can only survive if it joins up with itself smoothly instead of cancelling itself out, so the circumference has to be a whole number of wavelengths:

$$2\pi r = n\lambda = \frac{nh}{p} \implies pr = n\hbar. \quad (7)$$

This standing-wave requirement naturally produces Bohr's condition. The resulting permitted energy levels are:

$$E_n = -\frac{13.6 Z^2}{n^2} \text{ eV}, \quad (8)$$

and photons come off when electrons drop between levels. Whichever level they land on gives the series its name: Lyman, Balmer, Paschen, Brackett, Pfund. The app plots all five and labels them.

6.1 The reduced mass correction

Evaluating Equation 8 for $Z = 1$ yields $E_1 = -13.6057 \text{ eV}$ against the accepted -13.5984 eV , a 0.054% error. The discrepancy arises because the basic equation assumes the nucleus sits perfectly still while the electron goes round it. Both orbit their shared centre of mass, and the

proton, while heavy, is not infinitely heavy. The fix is to swap m_e for the reduced mass

$$\mu = \frac{m_e m_p}{m_e + m_p}, \quad (9)$$

and because E_n is directly proportional to mass, the fractional shift in energy is just the fractional difference between μ and m_e . The proton is about 1836 times heavier than the electron, so that difference is roughly $1/1837 \approx 0.054\%$. Redoing the calculation with μ yields -13.5983 eV , bringing the error down to 0.0007% . Both versions are in Table 1 for comparison.

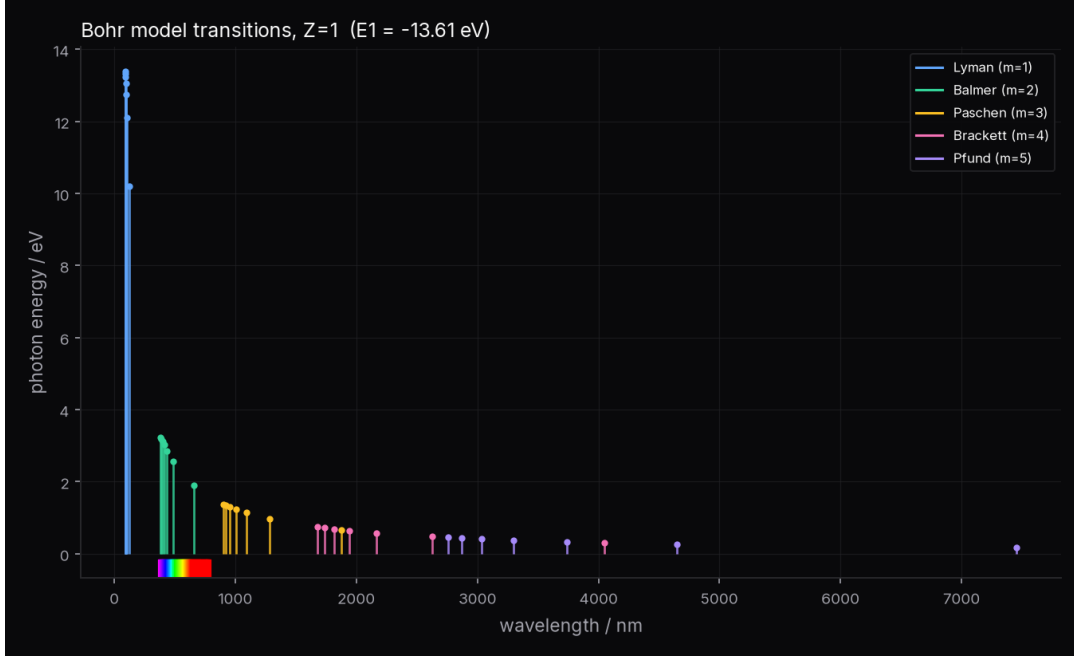


Figure 6: All five Bohr series for hydrogen, with a real visible-spectrum band drawn under the Balmer lines. The Lyman limit is at 91.1 nm.

7 Electron Diffraction

I thought of an electron as a tiny ball, and balls do not make diffraction rings when you fire them through a gap. So when electrons do make rings [10], they must be behaving as waves, with the same de Broglie wavelength from Task 5:

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2m_e eV}}. \quad (10)$$

Getting from a wavelength to a ring of a particular radius is basically three questions stacked on top of each other. This is why the pack gives three formulas rather than one. How long is the wavelength (de Broglie, above). At what angle does a wave that long interfere constructively off successive atomic layers (Bragg, $\sin \theta = n\lambda/2d$, the exact rule X-rays obey off any crystal). And where does a beam bent by that angle actually land on a flat screen a distance r away (pure geometry, $x = r \sin 2\phi$, doubled because reflecting off a lattice plane turns the beam by twice the glancing angle). Once I saw it as three questions rather than one formula, chaining them stopped looking arbitrary.

7.1 Avoiding a circular check

To avoid making the straight-line graph a trivial circular check, the recovery path shouldn't just perfectly reverse the generation path. I generated the ring radius x from the forward model and added 0.5 mm of Gaussian noise to mimic reading a real phosphor screen by eye. Then I recovered ϕ from the screen geometry $x = r \sin 2\phi$, and fitted $\sin \frac{1}{2}\phi$ against $1/\sqrt{V}$ through the origin. The gradient is $h/(2d\sqrt{2m_e e})$, allowing d to drop out naturally.

Through that noise it recovers $d_1 = 0.1245$ nm against a true 0.123 nm (1.2%) and $d_2 = 0.2122$ nm against a true 0.213 nm (0.37%), with $R^2 = 0.9946$ and 0.9856. This felt like a much more meaningful check.

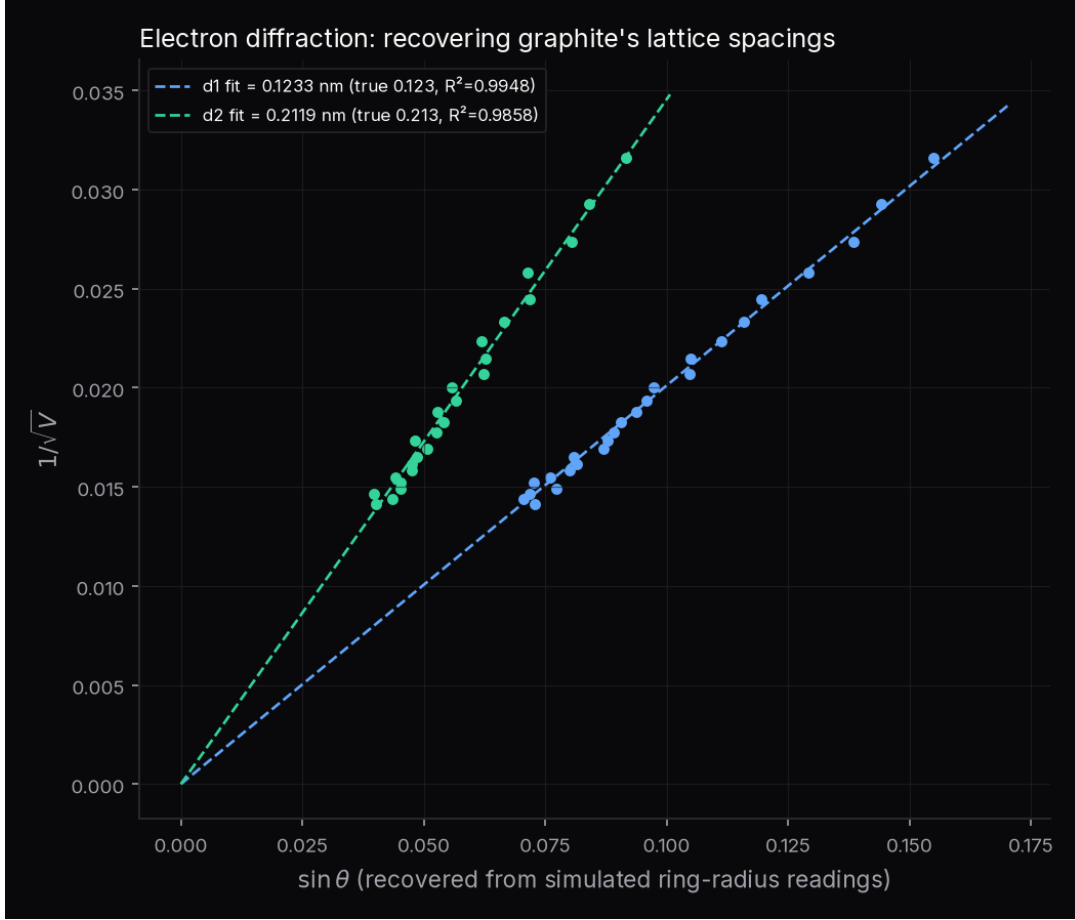


Figure 7: $1/\sqrt{V}$ against $\sin \frac{1}{2}\phi$, with ϕ recovered from simulated ring radii rather than from the Bragg condition that generated them, fitted back to graphite's two spacings.

8 The Particle in a Box

Plot energy against quantum number and the probability densities $|\psi|^2$ against x inside the box, with a write-up showing the system obeys the uncertainty principle as the extension.

The pack gives the energy levels and wavefunctions for a particle in a 1D box:

$$E_n = \frac{h^2 n^2}{8ma^2}, \quad \psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right). \quad (11)$$

I plotted E_n as a discrete ladder (since n is an integer) and plotted the probability densities $|\psi_n|^2$ rather than the wavefunctions themselves.

8.1 Extension: the uncertainty principle

The brief asks for a demonstration that $\Delta x \Delta p \geq \frac{1}{2}\hbar$ which was pretty exhausting to get through because I had to lean heavily on the examples in problem sheet QM3 especially for the $\sin^2$ integrals and the momentum operators so I've just put the full messy derivation in Section 12. The final answer comes out to:

$$\Delta x \Delta p = \hbar \sqrt{\frac{n^2 \pi^2}{12} - \frac{1}{2}}, \quad (12)$$

which doesn't depend on the box width a or the mass m at all, since they both cancel out. For the ground state, it evaluates to $0.5679\hbar$, which satisfies the $0.5\hbar$ bound, and it grows from there as n increases.

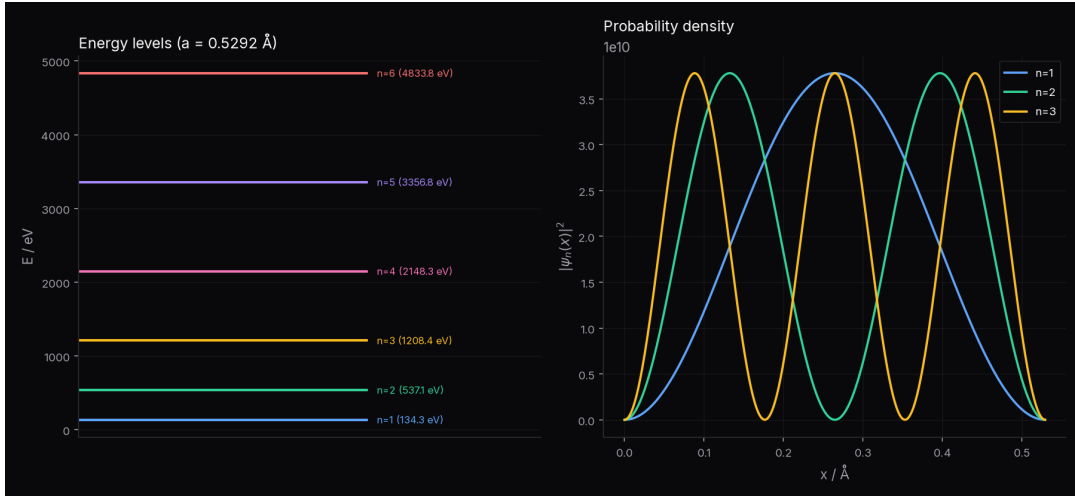


Figure 8: Left: the energy ladder, drawn as levels rather than the smooth curve I started with. Right: the probability densities, not the wavefunctions, for the first three levels.

9 Quantum Cryptography and the Bell Test

Build a visual calculator of the classical and quantum mismatch probabilities for detecting polarised entangled photons.

In this task, two entangled photons go to two detectors set at angles, each reporting one of two outcomes. A local hidden-variable model assumes that each photon carries a fixed polarisation from the moment it was created and the detectors just read it off. Bell's theorem [2] proves that no such model can reproduce what quantum mechanics predicts. The pack gives

$$P_{\text{classical}} = 1 - \cos^2 \theta \cos^2 \phi - \sin^2 \theta \sin^2 \phi, \quad (13)$$

$$P_{\text{quantum}} = \sin^2(\phi - \theta), \quad (14)$$

which agree at 0° and 90° and disagree everywhere in between, and when I plotted the resulting curves they perfectly matched the reference graph provided in the pack.

The quantum result depends only on $\phi - \theta$, the relative angle *between* the detectors, preserving rotational invariance. The classical model lacks this property. Put in $(\theta, \phi) = (0^\circ, 45^\circ)$ and then $(45^\circ, 90^\circ)$, the same 45° gap either way, and you get different answers, 0.5 and about 0.34. A photon carrying a fixed hidden property remembers some absolute direction from however it was made, so there is no reason it should care only about the relative angle the way a quantum one does. This asymmetry is intrinsic to the classical algebra, analyzed further in problem sheet QM3 (Section 14).

9.1 Extension: CHSH

The CHSH form [8] squashes all of this into one number. The combination of four correlations relies entirely on algebraic constraints rather than specific photon physics. In a hidden-variable world every photon already has an answer ready for both possible settings at its detector, even though only one gets used per trial. If you take one fixed set of those ready-made answers, each ± 1 , and evaluate

$$S = E(a, b) - E(a, b') + E(a', b) + E(a', b'). \quad (15)$$

The result is strictly ± 2 , whatever four values you plug in, because either $B(b) = B(b')$, in which case the first two terms cancel and the last two add to ± 2 , or $B(b) = -B(b')$ and it is the other way round. Now average over as many trials as you like, with any distribution over the hidden values you like: the average cannot exceed a bound that every single trial already obeyed. So $|S| \leq 2$.

Quantum outcomes are not drawn from a pre-decided menu—the core assumption required by the hidden-variable bound—allowing S to exceed 2. At the optimal angles I get $S = 2.8284$ for quantum mechanics and 1.4142 for the classical model. That 2.8284 is $2\sqrt{2}$, Tsirelson’s bound [7], the maximum permitted by quantum mechanics. Aspect’s experiment [1] closed the timing loophole, and the BB84 protocol [3] exploits these statistics for cryptographic security.

9.2 Simulation

To make sure I understood it, I also ran it as a simulation of individual photon pairs: Alice’s outcome is a coin flip, Bob’s is matched to hers with probability $\cos^2(\phi - \theta)$, and then the correlations are accumulated. A thousand trials per setting gave me $S \approx 2.85$, and Section 15 shows the estimator converging on $2\sqrt{2}$.

10 Compton Scattering

An X-ray photon bounces off an electron like a snooker ball and comes away at an angle with less energy. I expected this to be another place where the wave picture broke, the way it does in Task 4, and it mostly is not. All you actually have to swallow is that a photon carries momentum $p = h/\lambda$ despite having no mass, following from $E^2 = p^2c^2 + m^2c^4$ with $m = 0$. Accept that and conservation of energy and momentum does the rest. Compton’s own 1923 measurement [9] is what settled that photons carry momentum at all.

The pack gives the relativistic version of the shift, relativistic because the electron recoils at a decent fraction of c and you cannot get away with the simple form:

$$\frac{\Delta\lambda}{\lambda} = \frac{E}{m_e c^2} (1 - \cos \theta). \quad (16)$$

That $(1 - \cos \theta)$ has a picture behind it even if you never grind through the algebra. Draw the photon’s momentum before and after as two vectors of different length, since it lost energy, separated by the scattering angle θ . Whatever momentum the photon lost is exactly what the electron gained, by conservation, so the electron gets the vector difference between them. The length of that difference is a law-of-cosines problem, the same one you would use for any triangle, and that is where $\cos \theta$ comes in; combining it with energy conservation turns the geometry into Equation 16. The limits follow from the picture alone: a photon that does not scatter loses nothing, $1 - \cos 0^\circ = 0$, and one that comes straight back loses the most it possibly can, $1 - \cos 180^\circ = 2$. That maximum is $2\lambda_C$, twice the Compton wavelength, and it is checked in Table 1.

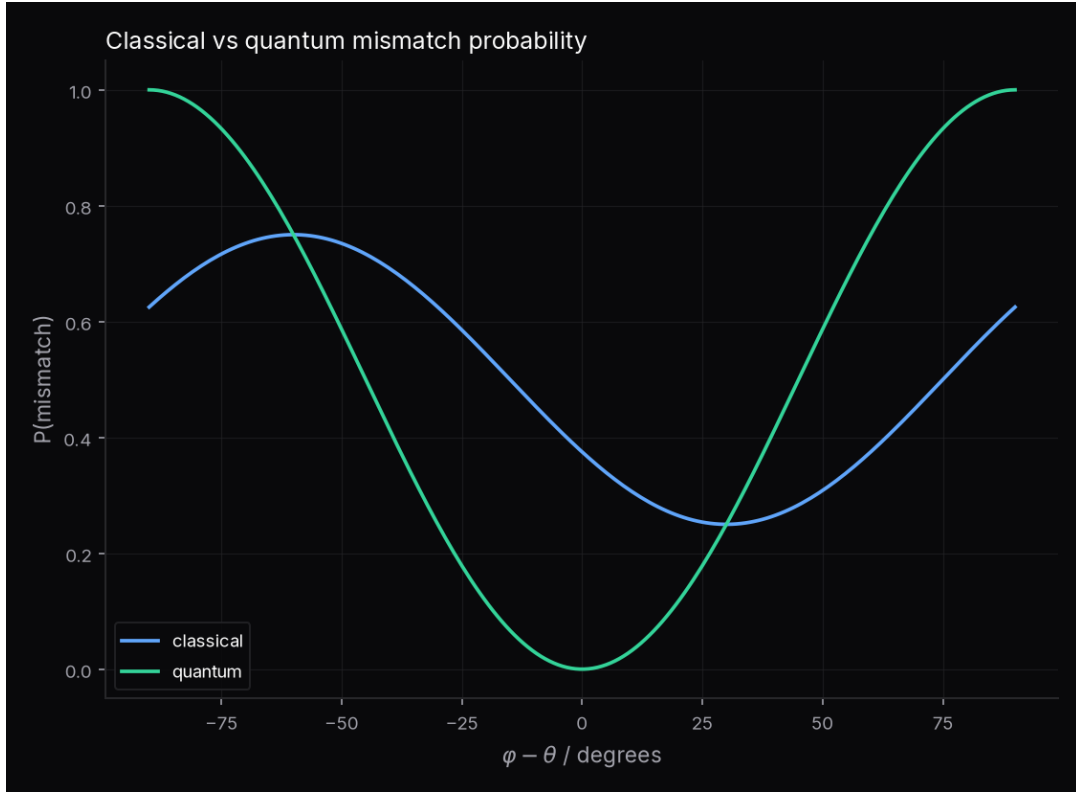


Figure 9: Classical against quantum mismatch probability. The CHSH numbers are reported live in the app rather than as a separate chart.

10.1 Visualizing the recoil

The extension draws the momentum arrows dynamically based on the slider. It made it much easier to spot things like how, at a scattering angle of 90° , the recoil angle drops from 42.3° at 50 keV to 18.7° at 1000 keV. Higher-energy photons drive the electron further forward. Playing with the visualization made this relationship much more intuitive than just staring at the data tables.

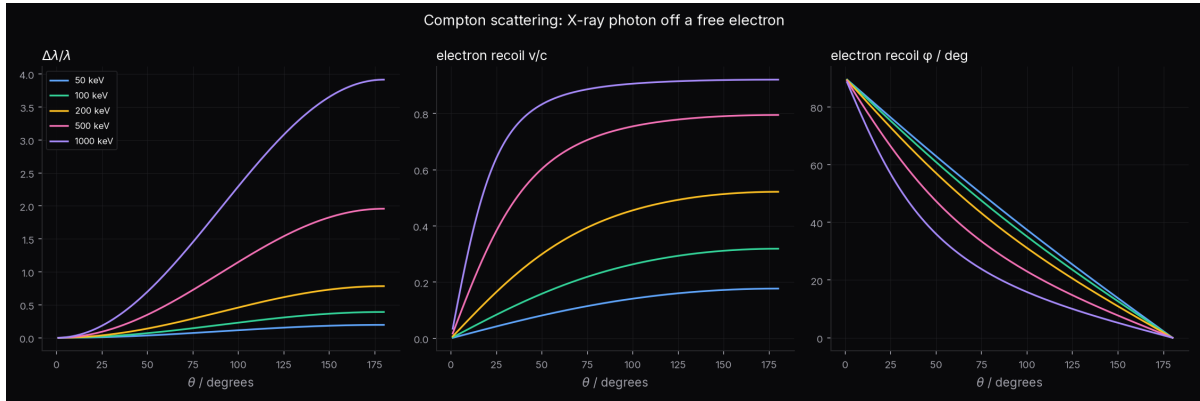


Figure 10: Fractional wavelength shift, recoil speed and recoil angle against scattering angle, for photon energies from 50 keV to 1 MeV.

11 Hydrogenic Orbitals

Plot 2D slices and 3D visualisations of $|\psi|^2$ for an electron in a hydrogenic atom given n , l and m , using the solutions provided in the pack.

The Schrödinger equation [22] separates into a radial part and an angular part:

$$\psi_{nlm}(r, \theta, \phi) = R(r, n, l) \cdot \Omega(\theta, \phi, l, m). \quad (17)$$

which makes sense as two separate questions once you look at it that way. The angular part Ω only answers “what shape is the cloud”, and those s, p, d and f lobe patterns are the real spherical harmonics, turning up whenever you solve a wave equation on the surface of a sphere. They are the same maths behind the vibration patterns of a struck drum skin. These functions are completely independent of the radial part, so they apply equally to the particle in a spherical box or the 3D harmonic oscillator, since there is nothing about these mathematical functions that is inherently specific to atoms. The radial part R only answers “how far out does it reach and how many shells does it have”.

To be honest I just plugged the Legendre and Laguerre polynomials straight into the code because the math behind them is well beyond what I’ve learned but you can sort of see how they trade off in the pictures where pushing n up gives you more shells and pushing l up makes the shape more complicated while the shells disappear.

11.1 Rendering challenges

I rendered $|\psi|^2$ as a 3D point cloud by rejection sampling: throwing random points into a box around the nucleus, keeping each with a probability proportional to $|\psi|^2$, and coloring by the sign of ψ .

Initially, my default camera view was configured to look straight down the z axis. Since every $m = 0$ orbital is symmetric about z , the s, p_z and d_{z^2} states all looked like featureless round blobs until I moved the camera to a three-quarter oblique position.

The 2D slice through $z = 0$ was also slightly confusing at first, because for $2p_z$, an orbital antisymmetric about that plane, the slice is just identically blank. I eventually added a note in the app (and later in Section 13) to explicitly state when an orbital has a node across the whole slice plane, so users don’t think the renderer is broken.

12 The Uncertainty Principle for a Particle in a Box

This is the derivation behind the PASS badge in Task 7, and it is also question 2 of problem sheet QM3. I follow that sheet’s route closely, so I am not claiming the method as mine, but I have written out the reasoning behind each step rather than just quoting results, and checked every number against the app’s live table. The principle itself is Heisenberg’s [15]; what is shown here is that this particular system obeys it at every level.

Throughout I use the normalised eigenfunctions of Equation 11, $\psi_n(x) = \sqrt{2/a} \sin(n\pi x/a)$ on $0 \leq x \leq a$.

12.1 Position uncertainty

You get $\langle x \rangle = a/2$ from symmetry without integrating anything. Reflect the box about its midpoint, $x \rightarrow a - x$, and $\sin^2(n\pi x/a)$ maps exactly onto itself for any whole n , so the density near the left wall is a mirror image of the density the same distance from the right. Anything

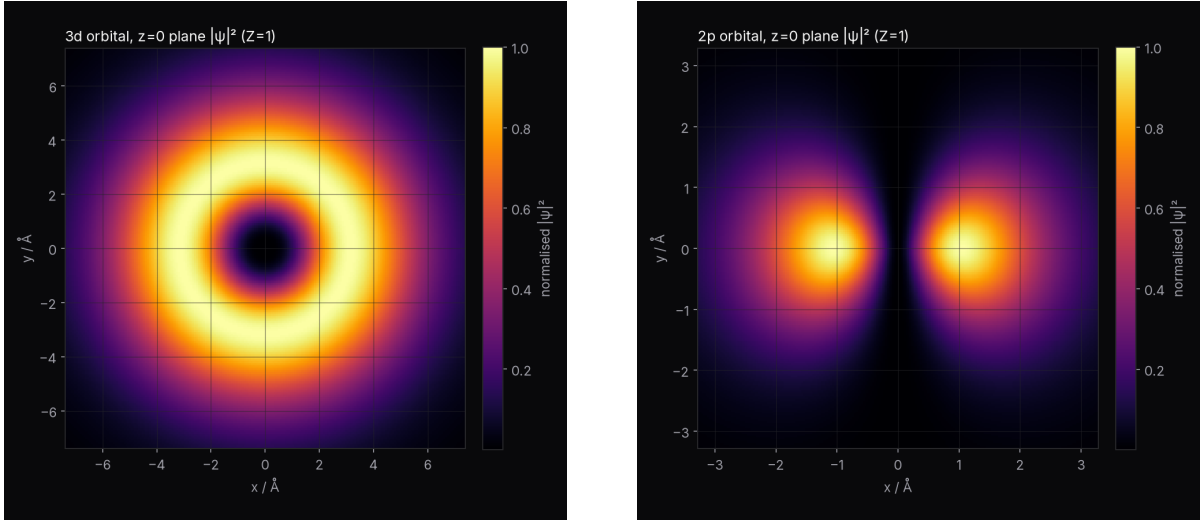


Figure 11: Left: the $3d$ orbital, $z = 0$ slice, as a $|\psi|^2$ heatmap. Right: the $2p$ ($m = \pm 1$) slice, which unlike $2p_z$ does have structure in the equatorial plane.

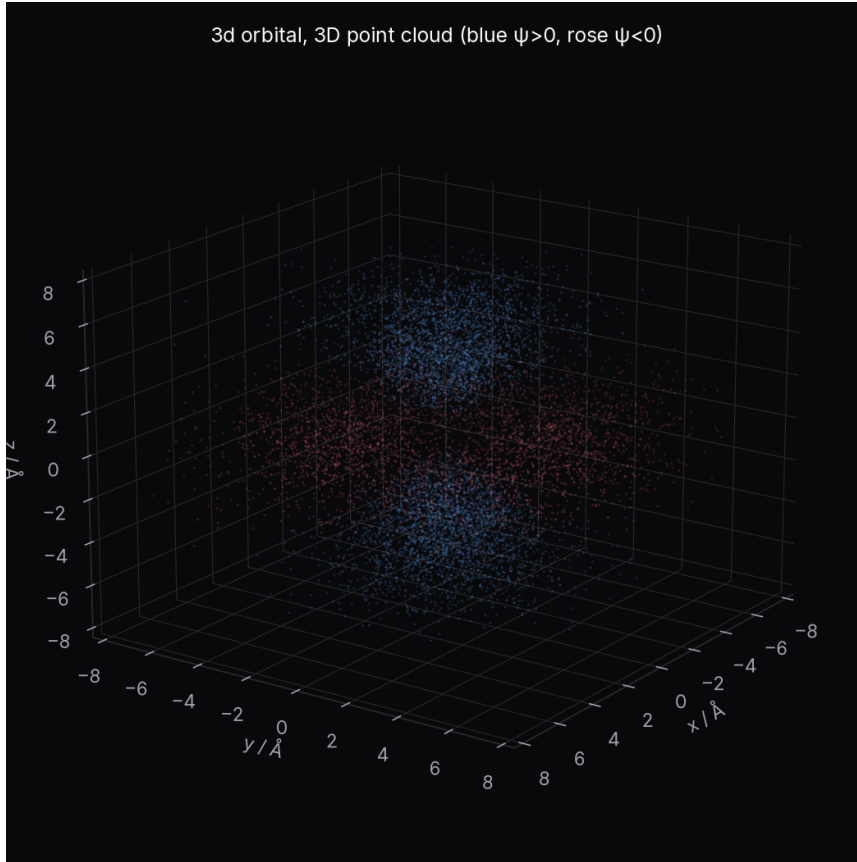


Figure 12: The same $3d$ orbital as a 3D point cloud, coloured by the sign of ψ .

that is a perfect mirror image of itself about $a/2$ has to average to $a/2$, the same reason a symmetric object's centre of mass sits on its own axis of symmetry.

The second moment is not free in the same way and needs an actual integral. Writing $\sin^2(n\pi x/a) = \frac{1}{2}(1 - \cos(2n\pi x/a))$ separates the integrand into a polynomial term and a term in $x^2 \cos(2n\pi x/a)$, and the second of those has to be integrated by parts twice, which is the actual grind. It gives

$$\langle x^2 \rangle = \int_0^a x^2 |\psi_n(x)|^2 dx = a^2 \left(\frac{1}{3} - \frac{1}{2n^2\pi^2} \right), \quad (18)$$

and hence

$$(\Delta x)^2 = \langle x^2 \rangle - \langle x \rangle^2 = a^2 \left(\frac{1}{12} - \frac{1}{2n^2\pi^2} \right). \quad (19)$$

12.2 Momentum uncertainty

$\langle p \rangle = 0$, again without integrating. A standing wave is not actually one travelling wave; it is a rightward-moving wave and a leftward-moving wave of equal size added together, which is what makes the fixed non-moving pattern you actually see rather than something that drifts. Exactly half of it goes each way, so the average momentum cancels.

For $\langle p^2 \rangle$ the sheet goes through the energy instead of the momentum operator. This makes sense once you notice there is no potential energy anywhere inside the box. The particle is free between the walls, so all its energy is kinetic, and kinetic energy is $p^2/2m$ for anything. So $\langle p^2 \rangle/2m = \langle E \rangle = E_n$ exactly, and with $E_n = n^2\pi^2\hbar^2/(2ma^2)$,

$$\langle p^2 \rangle = 2mE_n = \frac{n^2\pi^2\hbar^2}{a^2} \implies \Delta p = \frac{n\pi\hbar}{a}. \quad (20)$$

12.3 The product

Combining Equations 19 and 20,

$$\Delta x \Delta p = a \sqrt{\frac{1}{12} - \frac{1}{2n^2\pi^2}} \cdot \frac{n\pi\hbar}{a} = \hbar \sqrt{\frac{n^2\pi^2}{12} - \frac{1}{2}} = \frac{\hbar}{2} \sqrt{\frac{n^2\pi^2}{3} - 2}. \quad (21)$$

Both the box width a and the mass m cancel, a result I did not expect before doing the algebra: the uncertainty product here is a pure number times $\hbar$, depending only on which level you are in.

The bracket only grows with n , so the ground state is the tightest it ever gets, closest to the bound and never crossing it:

$$n = 1 : \quad \Delta x \Delta p = \hbar \sqrt{\frac{\pi^2}{12} - \frac{1}{2}} \approx 0.5679 \hbar > \frac{1}{2} \hbar. \quad (22)$$

So it holds at every level, with $n = 1$ the closest call, and it never gets close again. The app confirms that numerically for $n = 1$ to 6, and the values are plotted in Figure 17.

13 Hybrid and Molecular Orbitals

The brief's third extension opportunity asks for combinations of hydrogenic orbitals that approximate multi-electron atoms and multi-atom molecules, and separately for an animated

morph between different m values at fixed n and l . Those sound like three projects. They are one, and the reason is the single most useful thing I learnt doing this challenge.

The Schrödinger equation is linear. If ψ_1 and ψ_2 both solve it at the same energy, so does any weighted sum $c_1\psi_1 + c_2\psi_2$. So everything in this section is

$$\psi(\mathbf{r}) = \sum_i c_i \psi_{n_i l_i m_i}(\mathbf{r} - \mathbf{R}_i), \quad (23)$$

and $|\psi|^2$ is rendered by exactly the code Task 10 already had. Hybrid orbitals are Equation 23 with all $\mathbf{R}_i = 0$; molecular orbitals put the terms on different nuclei; the morph makes the c_i functions of a parameter. No new solver, no new renderer, three extensions.

13.1 Hybrid orbitals, and where bond angles come from

Mixing the one $2s$ with the three $2p$ orbitals on a single atom gives the standard hybrid sets, an idea due to Pauling [19]. Using the real (tesseral) orbitals from Task 10, where $m = 0, +1, -1$ are p_z, p_x and p_y , the tetrahedral set is

$$h_{1,2,3,4} = \frac{1}{2}(2s \pm 2p_x \pm 2p_y \pm 2p_z)$$

with an even number of minus signs in each. The geometry emerges naturally. Taking those superpositions is just the algebra required to get states that are orthogonal and point equally away from each other. The direction a hybrid points is read straight off its p coefficients, because $c_x p_x + c_y p_y + c_z p_z$ is just a single p orbital aligned along (c_x, c_y, c_z) . Taking the angle between the first two lobes of each set gives 180.00° , 120.00° and 109.47° , and those come out of the coefficient table, not out of a lookup. Against measured bond angles, sp matches acetylene's 180° and sp^3 matches methane's 109.5° ; sp^2 predicts 120° where ethene measures 117° , and the 3° gap is real physics the model omits, since the two C–H bonds and one C=C bond around that carbon are not actually equivalent.

As a check that the coefficient tables are right rather than merely plausible, treating the four atomic orbitals as an orthonormal basis makes each hybrid a 4-vector, and the Gram matrix of each set should be the identity. It is, to 2.5×10^{-16} for all three sets.

Getting a useful figure out of this took three attempts, and both of the failures taught me something. Summing $|\psi|^2$ over a complete hybrid set is exactly spherically symmetric, because the sum collapses to $|2s|^2 + |2p_x|^2 + |2p_y|^2 + |2p_z|^2$; the picture is a featureless ball, indicating that a filled shell has no preferred direction. Plotting a planar slice instead cannot show sp^3 at all, since those four lobes point along $(\pm 1, \pm 1, \pm 1)$ and no single plane contains more than two of them. Figure 13 therefore plots the angular part of each lobe separately as a polar surface, which is where all the directional information lives.

13.2 Molecular orbitals by LCAO

Putting the atomic orbitals on two different nuclei a distance R apart, still with Equation 23, gives molecular orbitals. Adding two $1s$ orbitals piles probability density into the region between the nuclei, where it is attracted to both at once: that is a bond. Subtracting them forces $\psi = 0$ on the plane midway between, so the electron is excluded from exactly the place that would have held the molecule together.

The normalisation is the part that says something. The two orbitals are not orthogonal, and their overlap

$$S(R) = e^{-\rho} \left(1 + \rho + \frac{1}{3}\rho^2 \right), \quad \rho = R/a_0,$$

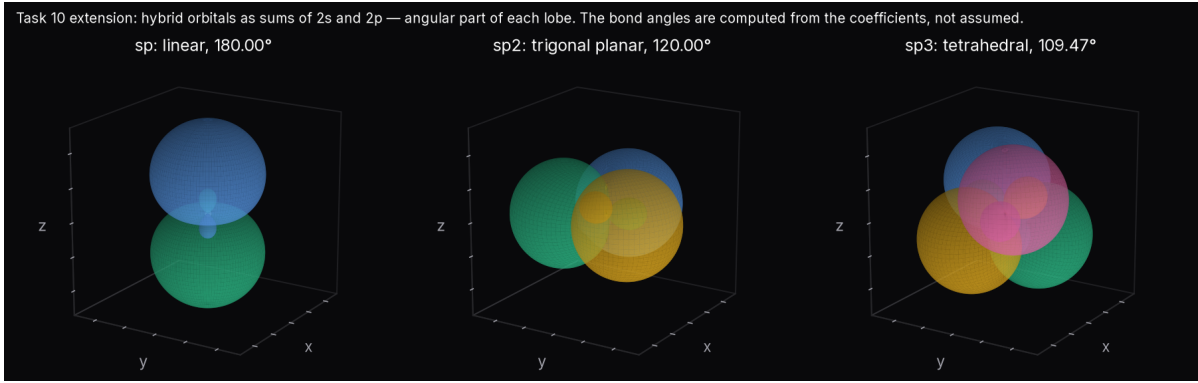


Figure 13: Hybrid orbitals as sums of $2s$ and $2p$, one lobe per colour. The angles in the titles are computed from the coefficients of each combination rather than assumed.

gives normalisation constants $N_{\pm} = 1/\sqrt{2(1 \pm S)}$. At the H_2 bond length $R = 0.741 \text{ \AA}$ this is $S = 0.7531$, so $N_+ = 0.5341$ and $N_- = 1.4230$. The two are not symmetric about 1, and that asymmetry is why the antibonding level is pushed up further than the bonding level is pushed down, which in turn is why He_2 does not exist: filling both levels is a net loss. I checked the closed form against a brute-force three-dimensional quadrature of $\langle 1s_A | 1s_B \rangle$ at four separations and they agree to 2×10^{-6} , which is the quadrature's own grid error.

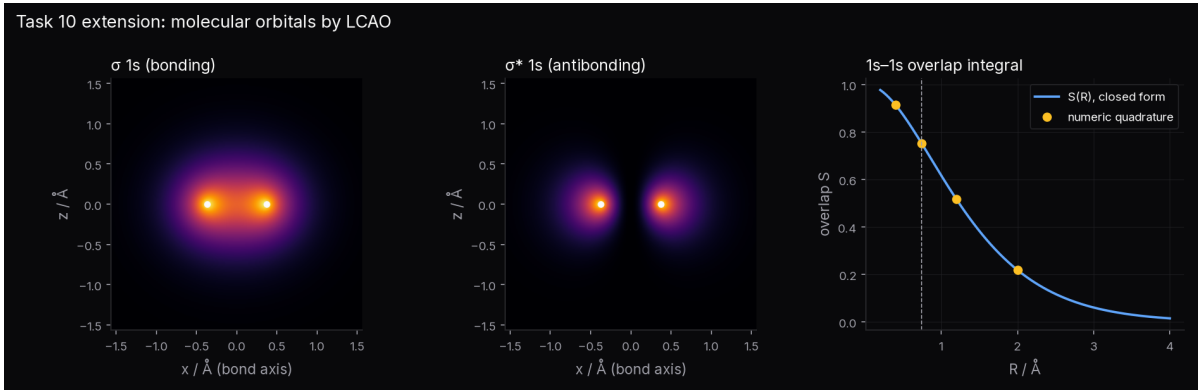


Figure 14: Bonding and antibonding σ orbitals from two $1s$ atomic orbitals at the H_2 bond length, and the overlap integral against separation with numerically integrated check points.

13.3 Morphing between m states

All $2l + 1$ values of m share an energy, so

$$\psi(\alpha) = \cos \alpha \psi_{nlm_A} + \sin \alpha \psi_{nlm_B}$$

is an exact solution at every α , not an interpolation between two solutions. The app animates α continuously.

Two things came out of building it. The first is a genuine physical result I had not seen coming: when $m_B = -m_A$ the morph is not a shape change at all. The real harmonics carry $\cos(m\phi)$ and $\sin(m\phi)$, and

$$\cos \alpha \cos(m\phi) + \sin \alpha \sin(m\phi) = \cos(m\phi - \alpha),$$

so the density is rigidly *rotated* about z by α/m and nothing deforms. I verified this numerically by comparing the $m = 2 \rightarrow m = -2$ superposition at $\alpha = 0.7$ against the $m = 2$ density

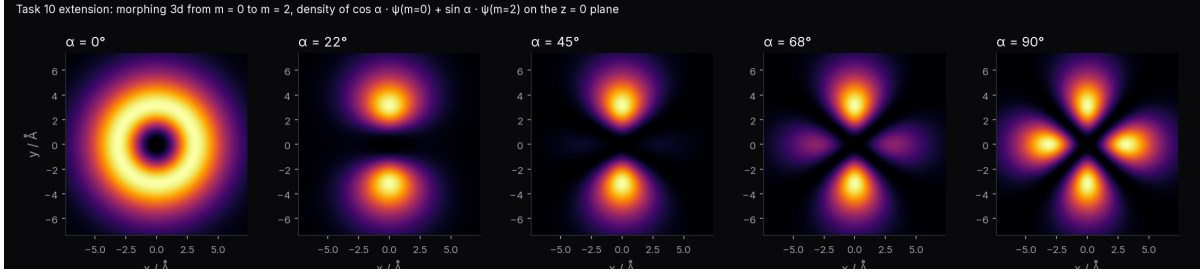


Figure 15: Morphing $3d$ from $m = 0$ to $m = 2$ on the $z = 0$ plane. Every frame is an exact solution of the Schrödinger equation at the same energy, -1.511 eV.

evaluated on a grid rotated by $\alpha/2 = 20.1^\circ$: they agree to 1.2×10^{-15} . Only morphs between different $|m|$ change the shape, because only those have different ϕ dependence that does not wash out. Figure 15 shows $3d$, $m = 0 \rightarrow m = 2$, deforming from a ring into four lobes.

The second is a performance problem with a tidy answer. Rejection-sampling a fresh 30 000-point cloud every frame is far too slow to animate. But by Cauchy–Schwarz,

$$(\cos \alpha \psi_A + \sin \alpha \psi_B)^2 \leq \psi_A^2 + \psi_B^2$$

for *every* α simultaneously, so $\psi_A^2 + \psi_B^2$ is an envelope that bounds the density at all morph angles at once. Sampling once against that envelope and caching both ψ_A and ψ_B at each accepted point means a frame only has to reweight points it already has, by $w(\alpha) = (\cos \alpha \psi_A + \sin \alpha \psi_B)^2 / (\psi_A^2 + \psi_B^2) \in [0, 1]$. That is arithmetic over two cached arrays, with no wavefunction evaluations in the animation loop at all, and it runs at display rate.

The endpoints are the check: at $\alpha = 0$ the rendered density must be exactly $|\psi_{m_A}|^2$ and at $\alpha = \pi/2$ exactly $|\psi_{m_B}|^2$. Computed independently in Python, the differences are 0 and 8×10^{-17} .

One practical detail worth recording, because it looks like a bug and is not: a real spherical harmonic vanishes on the whole $z = 0$ plane whenever $l + |m|$ is odd. Morphing $3d$ from $m = 0$ to $m = 1$ therefore gives a completely black 2D panel while the 3D cloud animates normally. The app spots that case and says so, rather than looking like it has failed.

14 Problem Sheets QM1–QM3

The brief lists answering the Quantum Mechanics problem sheets and writing them up as an extension opportunity, and notes that many of the questions link back to the challenge tasks. That turned out to be true in a stronger sense than I expected: three of these questions are the same physics as Tasks 4, 6 and 8 (Sections 5, 7 and 9) coming at it from the other end, and two of them changed how I had read the task briefs. This section covers sheets QM1, QM2 and QM3 in full. The Nuclear, Particles and Radioactivity sheets are not attempted here.

Every number below comes out of `project/src/problem_sheets.py` rather than being typed in by hand, so this text and the code cannot drift apart. Each sheet prints its own values for the physical constants and they are not identical between sheets ($h = 6.63 \times 10^{-34}$ Js on QM1 and QM2, 6.626×10^{-34} Js on QM3), so the code carries one constant set per sheet and uses the sheet’s own numbers for that sheet’s questions. Where my answer needed correcting against the published solutions I say so and explain what I had misread, because those were the places I learnt the most.

14.1 QM1: scales, radiation and the photoelectric effect

Q1. An atom is about 10^{-10} m across and a nucleus about 10^{-15} m, a ratio of 10^5 . The marble question is the memorable one. A 3.6 cm marble holds roughly $(3.6 \times 10^{-2}/2 \times 10^{-10})^3 \approx 5.8 \times 10^{24}$ atoms, and an Earth-sized container holds roughly $(1.28 \times 10^7/3.6 \times 10^{-2})^3 \approx 4.5 \times 10^{25}$ marbles. Those are the same to within a factor of eight; for an order-of-magnitude estimate, this places a marble almost exactly at the geometric mean of an atom and the Earth.

Q3. Treating the filament as a black body with area $A = Lw = 2.8 \times 10^{-5} \text{ m}^2$ and $P = \sigma AT^4$ gives $T = 2817 \text{ K}$, i.e. 2544°C , comfortably below tungsten's 3422°C melting point. The assumption doing the work is that all 100 W leaves as radiation and that emissivity is 1. It is worth noting that a real strip radiates from *both* faces, doubling A and dropping T to 2096°C , a 450°C difference; the sheet's solution uses the single-face area, so that is what I have quoted. Wien's law at 2817 K puts the peak at 1029 nm, in the infrared, which is the real answer to why an incandescent bulb is such a bad lamp: most of the 100 W never becomes visible light at all.

Stefan's law on the Sun ($R = 6.955 \times 10^8 \text{ m}$, $T = 5778 \text{ K}$) gives $L_\odot = 3.841 \times 10^{26} \text{ W}$, 0.35% above the accepted $3.828 \times 10^{26} \text{ W}$. At Mars ($2.279 \times 10^{11} \text{ m}$) that is 588.6 W m^{-2} , so a 20%-efficient panel delivering 100 W needs 0.850 m^2 , a square of side 0.92 m. Skin at 34°C peaks at 9442 nm, deep infrared, which is what thermal cameras see. The Sun peaks at 502 nm. For the red giant, constant luminosity means R^2T^4 is fixed, so $R \propto T^{-2}$: with $\lambda_{\text{max}} = 680 \text{ nm}$ giving $T = 4265 \text{ K}$, the radius grows by $(5778/4265)^2 = 1.84$.

Q4. Silver's 4.3 eV work function sets a threshold at 288.9 nm. At 200 nm each photon carries 6.211 eV, so photoelectrons leave with up to 1.911 eV ($3.058 \times 10^{-19} \text{ J}$) and the stopping voltage is 1.911 V. A $1.00 \mu\text{W}$ beam delivers 1.006×10^{12} photons per second, which at one electron per photon is a photocurrent of $0.161 \mu\text{A}$. This is Task 4 with the numbers run backwards, and it is a good check on that model: the app's apparatus reproduces all five of these.

Q5. The relativistic cutoff is the useful part. Requiring $v < 0.01c$ gives $V < m(0.01c)^2/2e = 25.6 \text{ V}$, so a classical treatment is only safe below about 25 volts, and the challenge's own 1–5 kV electron-diffraction range is well outside it. At 5 kV, $v/c \approx 0.14$: not catastrophic, but not negligible either, and Task 6 uses the classical de Broglie formula throughout.

The diffraction part is where I had to correct myself, and it is worth setting out because it changes how Task 6 should be read. The sheet's solution uses the grating condition $d \sin \theta = n\lambda$ with θ measured directly from the beam, so a first peak at 10° gives $\lambda = d \sin 10^\circ = 2.136 \times 10^{-11} \text{ m}$ and $V = h^2/(2m_e e \lambda^2) = 3306 \text{ V}$. I had originally used the Bragg condition $2d \sin \theta_B = n\lambda$ with θ_B the glancing angle and $2\theta_B$ the beam deviation, giving $\theta_B = 5^\circ$ and $V = 3281 \text{ V}$. The two agree to 0.8% here, purely because $2 \sin 5^\circ$ and $\sin 10^\circ$ differ by only 0.4%, so the accelerating voltage does not discriminate between them at all.

The number of orders does. Under the grating reading $\lambda/d = \sin 10^\circ = 0.1736$, so $n_{\text{max}} = 5$ and the peaks sit at 10.00° , 20.32° , 31.40° , 43.99° and 60.25° ; under the Bragg reading there are 11. Figure 16 shows both. The challenge pack's Task 6 asks for a $1/\sqrt{V}$ against $\sin \frac{1}{2}\phi$ straight line, which is the Bragg convention with ϕ the beam deviation, so the two documents are not using the same convention as each other. Neither is wrong; they are describing the geometry differently, and the accelerating voltage happens to hide the difference. (The published solutions print $\lambda/d = 0.1174$ at this point, but their own listed angles require 0.1736, and $1/0.1736 = 5.76$ is what gives their stated $n_{\text{max}} = 5$, so I have taken 0.1174 to be a slip.)

Q6. The Balmer formula $\lambda = 91.13 \text{ nm} / (m^{-2} - n^{-2})$ gives Lyman $3 \rightarrow 1$ at 102.5 nm (ultraviolet), Balmer $4 \rightarrow 2$ at 486.0 nm (visible, the blue-green line), and Pfund $7 \rightarrow 5$ at 4651 nm (infrared). These are the same three lines Task 5 draws, and they land on top of the app's stems.

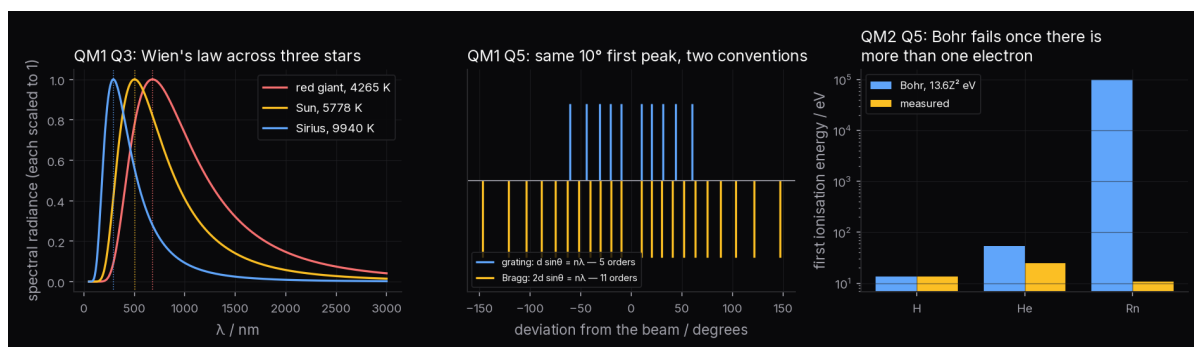


Figure 16: Left: Wien's law for the three stars these sheets ask about, each spectrum scaled to its own peak, dotted lines at $\lambda_{\max}$. Middle: QM1 Q5 under the two conventions, mirrored about the axis. Both put the first peak at 10° by construction; they disagree about how many orders exist. Right: the Bohr model against measured first ionisation energies, on a log scale, showing the model holding for hydrogen and failing by four orders of magnitude for radon.

14.2 QM2: photons, relativity and stars

Q1. The caesium standard's 9 192 631 770 Hz corresponds to $\lambda = 3.26$ cm, a microwave, and to an energy of 3.81×10^{-5} eV. That is around a millionth of a typical electronic transition and around 10^{11} times smaller than a nuclear one, so it is neither: it is a hyperfine splitting of the electronic ground state.

Q2. A 300 nm threshold gives a work function of 4.141 eV. The second part is the one I got wrong first time. "The photo-electron current constitutes 10% of the incident power" means 10 W of the 100 W ends up as photoelectron kinetic energy, and each electron carries $hc/\lambda - W = 3.313 \times 10^{-19}$ J, not the whole photon energy. That gives 3.019×10^{19} electrons per second and a current of 4.83 A. My first reading, that 10% of the incident *photons* produce electrons, gives 1.61 A. The wording supports the first one; I had swapped in the more familiar idea of quantum efficiency for what was actually written.

Q3. A water molecule modelled as 10 protons and 8 neutrons has $m = 3.013 \times 10^{-26}$ kg, and at $E = \frac{3}{2}k_B T = 6.21 \times 10^{-21}$ J its de Broglie wavelength is 34.3 pm. That is about a tenth of the molecule's own size, which is the point: the wave nature is there but far too small to matter, so water at room temperature is safely classical.

Q4. An electron at $0.5c$ needs 63.96 kV classically but 79.16 kV relativistically, so the classical formula is low by 19%. The wavelengths are 4.856 pm and 4.205 pm. Feeding the relativistic wavelength into a 2° deflection gives a spacing of 0.120 nm, which is within 3% of graphite's 0.123 nm layer spacing from Task 6, and the classical wavelength would have given 0.139 nm and missed it by 13%. This is a concrete case where the relativistic correction is the difference between identifying the material and not.

Q5. The Bohr model predicts ionisation energies of 13.6 eV, 54.4 eV and 100.6 keV for H, He and Rn, at wavelengths 91.35 nm, 22.84 nm and 0.0124 nm. Measured first ionisation energies are 13.62 eV, 24.62 eV and 10.76 eV, at 91.24 nm, 50.46 nm and 115.4 nm. Hydrogen agrees to 0.1%; helium is out by a factor of 2.2; radon by a factor of 9300.

The reason is that the two quantities are not the same thing. Bohr's $13.6Z^2$ eV is the energy to remove the *only* electron from a bare nucleus of charge Z , so for helium it is really the ionisation energy of He^+ , not of neutral He. A measured first ionisation energy is the energy to remove the outermost electron of a neutral atom, which is screened from the nucleus by all the others. For radon that electron sits in $n = 6$ and is screened by 85 others, so it sees an effective charge of order 1 rather than 86, and Z^2 scaling is meaningless. The model is not approximately wrong here; it is answering a different question.

Q6. At 1420 MHz each photon carries 9.41×10^{-25} J. Andromeda's 8×10^{27} W at 2×10^{22} m gives a flux of 1.59×10^{-18} W m $^{-2}$, and FAST's 1.96×10^5 m 2 dish collects 3.32×10^{11} photons per second. A DSLR at 100 W m $^{-2}$ for 1/500 s over 24×36 mm collects 4.35×10^{14} photons in one frame. FAST would need 1310 s, about 22 minutes, to accumulate as many photons as a single daylight snapshot.

Q7. The solar sail result is the elegant one. $N = IA\lambda/hc$ photons arrive each second, each reflecting with momentum change $2h/\lambda$, so

$$F = N \cdot \frac{2h}{\lambda} = \frac{IA\lambda}{hc} \cdot \frac{2h}{\lambda} = \frac{2IA}{c},$$

and the wavelength cancels completely. Momentum flux does not care how the energy is parcelled into photons. For $I = 1$ kW m $^{-2}$, $A = 100$ km 2 and $m = 10$ tonnes this is 667 N and $a = 0.0667$ m s $^{-2}$, reaching $c/100$ in 4.49×10^7 s, about 1.42 years.

Q8. From the 1361 W m $^{-2}$ solar constant, $L_\odot = 3.828 \times 10^{26}$ W. Over 10 Gyr that radiates 1.34×10^{27} kg of mass-energy, 0.068% of the Sun's mass. Sirius, at 9940 K and $1.711 R_\odot$, would deliver 35 074 W m $^{-2}$ at 1 AU, 26 times the solar constant, and has $L = 9.86 \times 10^{27}$ W = $25.8 L_\odot$ against a measured value of about $25 L_\odot$. Losing the same mass fraction, Sirius lasts 2.47×10^{16} s, roughly 0.78 billion years against the Sun's ten: eight per cent of the lifetime for twice the mass, which is the mass-luminosity relation in one line. The heating element in the last part draws 351 kW and boils a litre of water from 20 °C in 0.95 s.

14.3 QM3: uncertainty, confinement and entanglement

Q1(i)–(ii). An electron at $c/137$ has $\lambda = 3.324 \times 10^{-10}$ m. That number is not a coincidence: it is $2\pi a_0$ to four significant figures, i.e. exactly the circumference of the $n = 1$ Bohr orbit, which is the standing-wave condition $n\lambda = 2\pi r$ from the top of the sheet. The fine-structure constant $\alpha \approx 1/137$ is the ground-state orbital speed in units of c , so this question is the Bohr model in disguise. For an electron and a neutron of equal wavelength the momenta are equal, so speeds are in the ratio $m_n/m_e = 1839$, and since $E = p^2/2m$ the kinetic energies are in that same ratio.

Q1(iii). Taking $\Delta p \sim mc$ and $\Delta E \sim mc^2$ for the Z boson at 91.2 GeV/ c^2 gives a range $\Delta x = \hbar/2mc = 1.08 \times 10^{-18}$ m, about a thousandth of a proton radius, and a lifetime $\Delta t = \hbar/2mc^2 = 3.6 \times 10^{-27}$ s. The measured Z lifetime is about 2.6×10^{-25} s, so a one-line uncertainty estimate lands within two orders of magnitude of a quantity measured at LEP, which is about as much as an order-of-magnitude argument can promise.

Q1(vii). Neutrons on a lattice of spacing $s = 2.15 \times 10^{-10}$ m, with the first two maxima 30° apart. The two maxima meant are the central $n = 0$ peak and the $n = 1$ peak, so 30° is the position of the first order itself and $\lambda = s \sin 30^\circ = s/2 = 1.075 \times 10^{-10}$ m exactly. Then $E = 0.0708$ eV and $v = 3680$ m s $^{-1}$. Comparing with $\frac{3}{2}k_B T = 0.0385$ eV at room temperature shows these are essentially thermal neutrons, which is exactly why reactor neutrons are usable for diffraction without any acceleration stage.

Q1(viii)–(x). Silver again: $4.3+5.1 = 9.4$ eV photons are 131.9 nm ultraviolet. An electron of that same wavelength has only 8.6×10^{-5} eV, because $E \propto 1/\lambda^2$ for massive particles but $E \propto 1/\lambda$ for photons, and the closed form $E_e = (E + \phi)^2/2m_e c^2$ makes the ratio explicit. A microscope resolving 0.42 nm needs 8.53 eV electrons. A 5 MeV beta decay gives $\Delta t \geq 6.58 \times 10^{-23}$ s; using the relativistic momentum ($pc = 5.487$ MeV) rather than the classical one gives $\Delta x \geq 1.80 \times 10^{-14}$ m, a few nuclear radii, and $c \Delta t = 1.97 \times 10^{-14}$ m independently, which is the same answer by a different route.

Q2. This is Task 7. The parts already covered in Section 12 are the normalisation, $\langle x \rangle$,

$\langle x^2 \rangle$, $\langle p \rangle$, $\langle p^2 \rangle$ and the uncertainty product; the sheet's closed form

$$\Delta x \Delta p = \frac{\hbar}{2} \sqrt{\frac{n^2 \pi^2}{3} - 2}$$

is the same expression derived there, rearranged, and at $n = 1$ both give $0.5679\hbar$.

Part (ii) is new and is the best question on the sheet. Putting a proton in a box of width $a = 0.47$ fm, roughly a nuclear diameter, gives

$$E_1 = \frac{h^2}{8m_p a^2} = 927.2 \text{ MeV},$$

against a proton rest energy of 938.4 MeV, giving a ratio of 0.988. A confinement energy of order the rest energy means the non-relativistic Schrödinger equation used to derive it is not valid at this scale, so the calculation refutes its own assumptions, and it does so by almost exactly a factor of one, and this is a compact demonstration of why. The honest conclusion is that nucleons confined to nuclei are a relativistic problem.

Q3. Two detectors at θ and ϕ , and the mismatch probability computed two ways. Treating the measurements as independent gives

$$P = \cos^2 \theta \sin^2 \phi + \sin^2 \theta \cos^2 \phi,$$

and under the Copenhagen interpretation, where A's measurement collapses B's photon into A's outcome state, conditioning on both of A's outcomes gives

$$P = \cos^2 \theta \sin^2(\phi - \theta) + \sin^2 \theta \sin^2(\phi - \theta) = \sin^2(\phi - \theta).$$

These are Task 8's two formulas, derived rather than quoted, and the first is algebraically identical to the pack's $P_{\text{classical}} = 1 - \cos^2 \theta \cos^2 \phi - \sin^2 \theta \sin^2 \phi$, an equivalence the code asserts numerically.

The sign convention nearly caught me out. The sheet's figure puts the two detectors on *opposite* sides of the vertical, so in one signed convention $\theta = -45^\circ$ and $\phi = +30^\circ$ and they are 75° apart, not 15° . The independent-detector answer is 0.5 either way, because $\cos^2(\pm 45^\circ) = \sin^2(\pm 45^\circ)$, but the quantum answer is $\sin^2 75^\circ = 0.933$ one way and $\sin^2 15^\circ = 0.067$ the other. Those two are complements, so getting the sense wrong does not merely shift the answer, it inverts the conclusion from "quantum mechanics predicts far more mismatches than a hidden-variable model" to "far fewer". With the sheet's geometry the quantum prediction is 1.87 times the classical one, and Figure 17 shows where the two curves separate as a function of detector separation. This is the same gap that Task 8's CHSH extension turns into the single number $S = 2\sqrt{2}$.

15 Validation

Checking the models was probably the hardest part of the project because a lot of the time the code ran perfectly and drew a nice graph but the physics was still completely wrong. The only thing that reliably caught them was checking against something independent, so this section puts all of those checks in one place. Everything here is generated by `project/src/validation.py`; Table 1 is written directly by that script rather than transcribed, so the paper cannot fall out of step with the code.

I ended up sorting the validation checks into two different categories.

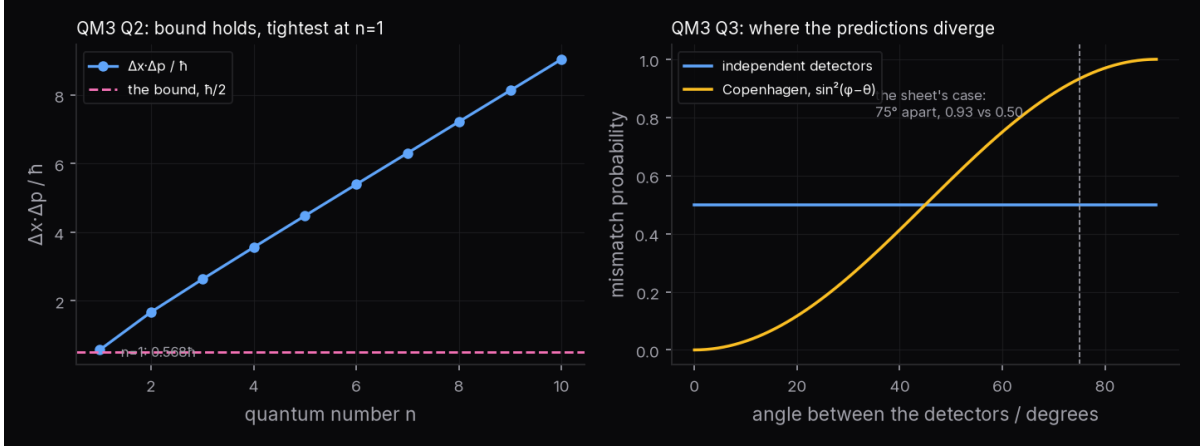


Figure 17: Left: the uncertainty product for the infinite well, in units of $\hbar$, against n . It never crosses the bound and is tightest at $n = 1$. Right: mismatch probability against the angle between the detectors, for independent measurements and under the Copenhagen interpretation. The sheet’s case sits at 75° .

15.1 Deterministic checks

These compare something I computed against something known independently: a CODATA constant [24], a closed-form integral, a published bond angle. They either agree or they have found a bug.

The most informative is the Stefan–Boltzmann check on Task 3. Numerically integrating Planck’s law over all wavelengths must return σT^4 , and σ appears nowhere in the Planck function, so recovering it tests the formula, the constants and the quadrature simultaneously. It agrees to better than 0.0003% across 3000–9940 K.

Two rows in the table are there because they caught something. The hydrogen ground state computed with the bare electron mass gives -13.6057 eV against the accepted -13.5984 eV, a 0.054% error, and using the reduced mass $\mu = m_e m_p / (m_e + m_p)$ instead brings it to 0.0007%. That factor is the electron and proton orbiting their common centre of mass rather than the proton sitting still, and it explained the whole discrepancy. It was just a piece of physics I had missed. The Task 6 entry is the other one. Recovering the graphite spacing by inverting my own forward formula on exact inputs reproduces d perfectly and proves nothing, so the ring radii there carry 0.5 mm of Gaussian noise, about what reading a phosphor screen by eye is worth. Through that, the fit still recovers 0.123 nm and 0.213 nm to 1.2% and 0.4%.

15.2 Stochastic checks

Tasks 1, 2 and 8 are Monte Carlo, and a single run disagreeing with theory by a few per cent is not evidence of anything at all. I spent a long time early on treating Task 1’s $\langle r^2 \rangle$ of 355 against a predicted 399 as a bug to be found. But I realised later that I was just taking a single run of a random variable and treating the answer as an exact measurement.

The right test is on the *rate* at which the error falls. An unbiased estimator with finite variance has error $\propto M^{-1/2}$ in the sample count M , and a biased one flattens out at some non-zero floor instead, so fitting the exponent β in error $\sim M^{-\beta}$ distinguishes the two. Sweeping M over three decades and fitting on log axes gives

$$\beta = 0.555 \quad (\text{Task 1}), \quad \beta = 0.466 \quad (\text{Task 8}),$$

both consistent with $1/2$, with no sign of a floor (Figure 18). At the largest sample counts the random walk is within 1.6% of Ns^2 and the Monte Carlo CHSH value is within 0.004 of $2\sqrt{2}$.

That is a much stronger statement than any single run could make, and it is the difference between "my number is close" and "my estimator is correct".

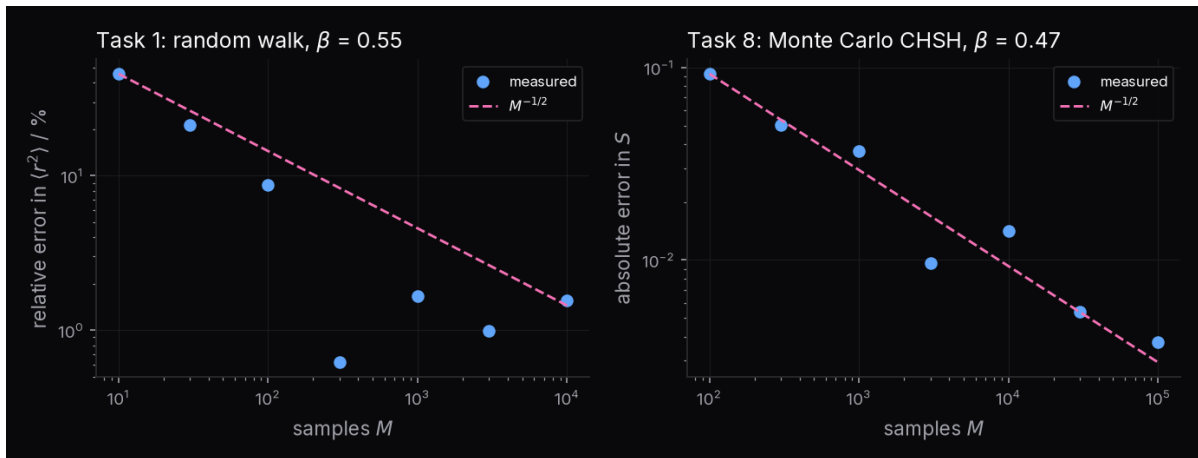


Figure 18: Error against sample count on log axes for the two Monte Carlo tasks, with an $M^{-1/2}$ reference line. Both track the reference slope rather than levelling off, indicating the estimators are unbiased.

15.3 Cross-implementation agreement

Everything here exists twice, once in TypeScript in the app and once in Python for these figures. That started out purely so this write-up would not be a folder of screenshots, but it turned into a useful debugging tool. The hydrogenic radial function is a hand-rolled Laguerre recurrence in TypeScript and `scipy.special` in Python; the LCAO overlap at the H_2 bond length comes out as $S = 0.7531$ with normalisations 0.5341 and 1.4230 in both, to every digit the app shows. Several of the silent bugs above first turned up as a disagreement in the fourth significant figure between the two.

Table 1: Deterministic checks, generated by `validation.py`. Percentage errors are against the reference column.

Task	Check	Computed	Reference	Error
3. Blackbody	$\pi \int B d\lambda$ vs σT^4 at 3000 K	4.59767×10^6	4.59768×10^6	$2.18 \times 10^{-4}\%$
	$\pi \int B d\lambda$ vs σT^4 at 5778 K	6.32650×10^7	6.32650×10^7	0%
	$\pi \int B d\lambda$ vs σT^4 at 9940 K	5.54114×10^8	5.54114×10^8	0%
5. Hydrogen (Bohr)	Bohr radius a_0 / m	5.29177×10^{-11}	5.29177×10^{-11}	$3.99 \times 10^{-5}\%$
	E_1 with reduced mass / eV (<i>reduced mass matters at the 0.05% level</i>)	-13.5983	-13.5984	$7.35 \times 10^{-4}\%$
	E_1 with electron mass / eV (<i>using m_e rather than μ is the whole error</i>)	-13.6057	-13.5984	0.05368%
9. Compton	Compton wavelength λ_C / m	2.42631×10^{-12}	2.42631×10^{-12}	$9.84 \times 10^{-6}\%$
	backscatter shift $2\lambda_C$ / m	4.85262×10^{-12}	4.85262×10^{-12}	0%
	511 keV backscatter / keV (<i>closed form $m_e c^2/3$</i>)	170.333	170.333	0%
7. Particle in a box	$\Delta x \Delta p / \hbar$ at $n = 1$ (<i>must exceed 0.5</i>)	0.567862	0.567862	0%
8. Bell / CHSH	S at the optimal angles (<i>Tsirelson bound</i>)	2.82843	2.82843	0%
10 ext. Hybrids	sp inter-lobe angle / degrees	180	180	0%
	sp2 inter-lobe angle / degrees	120	120	0%
	sp3 inter-lobe angle / degrees	109.471	109.471	0%
10 ext. LCAO	overlap S at $R = 0.741 \text{ \AA}$, closed form vs quadrature	0.753075	0.753077	$2.66 \times 10^{-4}\%$
	overlap S at $R = 1.2 \text{ \AA}$, closed form vs quadrature	0.516197	0.516199	$3.87 \times 10^{-4}\%$
4. Photoelectric	h from the stopping-voltage gradient / Js (<i>12 points, 20 mV noise</i>)	6.63964×10^{-34}	6.62607×10^{-34}	0.2048%
	work function from intercept / eV	2.31162	2.3	0.5052%
6. Electron diffraction	d from the ring-radius fit / nm (true 0.123) (<i>0.5 mm radius noise, $R^2 = 0.9946$</i>)	0.124468	0.123	1.193%
	d from the ring-radius fit / nm (true 0.213) (<i>0.5 mm radius noise, $R^2 = 0.9856$</i>)	0.21222	0.213	0.3662%

16 The Interactive Application

16.1 Introduction

The brief notes that constructing an app with a graphical user interface counts as extension work, and that an interactive website will particularly impress the judges. Rather than write ten separate scripts and a launcher, I built one app where each task is a slide of a keyboard-driven deck. Every number on screen is recomputed from the controls the moment you move them: the numbers are calculated on the fly and the figures are drawn directly in the browser. This section covers how it is built and how it deploys.

16.2 Libraries and technologies

The app is TypeScript [18] on React 19 [17], bundled by Vite [26]. Styling uses Tailwind CSS [23] with the shadcn/ui component primitives, providing the sliders, toggles and cards used by the controls. Two-dimensional line charts are drawn with Recharts [21]; the stem, scatter and geometry diagrams that no standard chart type fits are drawn as raw SVG. The three-dimensional orbital rendering uses three.js [6] through react-three-fiber, with the heatmaps drawn directly to an HTML canvas.

The Inter font is self-hosted rather than pulled from a font service, so after the page loads the app makes no external requests at all. That was deliberate: it has to work off a laptop with no internet, because that is a very plausible situation to end up demonstrating it in.

16.3 Architecture

Physics and presentation are kept completely apart. Every model sits in `src/lib/physics/` as a module that imports nothing from React and knows nothing about rendering, exporting plain functions of its physical parameters. The interface imports those and handles only controls, layout and drawing.

Table 2: Module layout of the application.

Path	Responsibility
<code>src/lib/physics/</code> <code>src/lib/palette.ts</code>	One module per task. Pure functions, no UI dependencies. Shared categorical colour palette, matched to the Python figures.
<code>src/slides/</code>	One component per task: controls, layout and choice of visualisation.
<code>src/components/deck/</code> <code>src/components/ui/</code>	Deck shell, slide chrome, charts, 3D scenes, canvas helpers. shadcn/ui primitives (slider, toggle, card, tabs).

That split is what made the build-it-twice approach of Section 15 practical. Because the physics modules are plain functions with no framework tangled into them, I can call them straight from a console and compare against Python without instantiating any interface at all, which is exactly how those cross-checks were run.

16.4 Functionality

Each slide presents one task, with a control panel on the right and one or two visualisations filling the remainder. Navigation is by arrow key, with **Home** and **End** jumping to the first and

last slide and **f** toggling fullscreen; clicking the left or right screen edge also navigates, as a fallback during a live demonstration.

Task 10 (Section 11) carries the extension work of Section 13 behind a mode selector:

- **Single** ψ_{nlm} renders one hydrogenic orbital, the base task.
- **Hybrid** renders the sp , sp^2 and sp^3 sets, either as all lobes at once or one at a time, and reports the inter-lobe angle computed from the coefficients against the measured bond angle of a representative molecule.
- **Molecular** renders bonding and antibonding σ and π orbitals with a variable internuclear separation, reporting the overlap integral $S(R)$ and both normalisation constants live.
- **m-morph** animates $\cos \alpha \psi_{m_A} + \sin \alpha \psi_{m_B}$ continuously, with the superposition coefficients and the energy displayed as it runs.

16.5 Implementation notes

Three problems here were solved in ways worth writing down.

WebGL context limits. Since each of those four modes originally mounted its own `three.js` canvas, I quickly ran into the browser limit on how many WebGL contexts can be active at the same time. If you flipped between the modes a bunch of times it would drop the oldest one and break the animation loop. It was super annoying to debug because there were no error messages in the console but I eventually figured out I had to hoist a single canvas component above the mode switch and just swap what was being drawn.

Sampling performance. The point clouds use rejection sampling, and I initially set the envelope to the true maximum density like the textbook says. However, this turned out to be a terrible idea for anything with a $2s$ orbital because the spike at the nucleus causes almost everything else to get rejected. The sp^3 set was taking ages to render and still looked completely bare. I fixed it by just using the 97th percentile as the envelope instead which basically just ignores a tiny sliver of volume right at the nucleus.

Animation speeds. Trying to resample the whole point cloud every single frame for the morph animation was way too slow to actually run in real time. Because of the Cauchy–Schwarz bound I mentioned in Section 13 I was able to just sample the cloud once and then reweight the points every frame instead of running the wavefunction evaluations inside the animation loop.

16.6 Deployment

The application is entirely static so it just runs as static files. `pnpm build` produces a `dist/` directory that can be served from any static host, and configuration files for both Vercel and Netlify are included in the repository, each setting the build command, the publish directory, a single-page-application rewrite and long-lived cache headers for the fingerprinted assets.

16.7 Summary and possible improvements

The app handles all ten tasks interactively, computing physics live. Keeping the physics modules completely separate from the UI framework ended up being really useful when adding the extension modes.

If I were to keep working on it, I'd probably look at the initial load time—the production bundle is 1.7 MB before compression, mostly because of `three.js`, so lazy-loading that library

only when it is needed for Task 10 would help significantly. Also, the point-cloud sampling still runs on the main thread, so a very large point count can cause a brief stutter when changing modes; moving that to a Web Worker is the obvious fix. Finally, adding URL routing so you can link directly to a specific slide would make it easier to reference from the write-up.

16.8 Links

- Application: <https://bpho.antwi.dev>
- Source: `webapp/` in the accompanying repository
- Python models, problem-sheet solver and validation suite: `project/src/`

17 Conclusion

This project involved building ten computational models, along with some extension work on the uncertainty principle, a Bell test simulation, superpositions of hydrogenic orbitals, and working through problem sheets QM1 to QM3.

17.1 A few surprises

The geometry of chemical bonding emerges organically from the orbital superpositions, reproducing the standard sp , sp^2 , and sp^3 angles (180.00° , 120.00° , and 109.47°) algebraically (Section 13).

Modeling a proton in a nucleus-sized box yields a non-relativistic ground-state energy of 927.2 MeV against a rest energy of 938.4 MeV (Section 14). A confinement energy this close to the rest energy explicitly demonstrates the breakdown of non-relativistic assumptions at the nuclear scale.

The challenge pack and the problem sheets employ different angular conventions for the electron-diffraction experiment: the Bragg condition versus the grating condition. While the predicted accelerating voltages differ by only 0.8%, the models diverge significantly in the number of predicted diffraction orders, five versus eleven (Section 14).

17.2 Takeaways from the project

The most persistent errors were those that produced visually plausible but physically incorrect results—such as a random walk sampling a square envelope, plotting discrete energy levels as a continuum, or neglecting the reduced mass. Identifying these required independent verification methods, such as numerically integrating Planck’s law to recover the Stefan-Boltzmann constant, or evaluating Monte Carlo convergence rates rather than single-run outputs.

17.3 Future work

There are a few things left unfinished. The Nuclear, Particles and Radioactivity problem sheets that came with the challenge are not attempted here, and doing them would round out Section 14. The LCAO work obvious next step given the hybrid orbitals are already sitting there as a basis. Furthermore, the Brownian motion simulation (Task 2) remains qualitative. Extracting a diffusion coefficient from the particle trajectory and comparing it against the Stokes–Einstein relation would provide the necessary quantitative verification.

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